Case Study: Ranking of Players in a tournament using Matrix Model

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1. Introduction

Modern competitive tournaments consist of several interdependent players, much like the sectors of a global economy. In this system, a single match result does not exist in isolation; instead, each player's performance serves as an input that affects the relative standing of every other participant.

The Economic Input-Output Model, originally developed by Wassily Leontief, provides a systematic mathematical framework to analyze these complex, inter-player relationships. By treating a tournament as a closed loop of "production and consumption," this model helps determine the total rating points required for each player to satisfy a given win-loss record, while mathematically accounting for the strength of schedule and intermediate difficulty within the competition.

Case Study :- Ranking Players In A Tournament Using Matrix Model.

→ Mathematical Modelling of the Tournaments Ranking System.

n : The number of players or terms in the tournament.

x_i = The total rating of player i.

d_i = The Base Performance of player i.

a_{ij} = The Interaction Coefficient.

The coefficients are arranged in a square matrix A.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Each element a_{ij} quantifies how much the ranking of player i depend on the ranking of player j.

The total rating must satisfy the sum of their Internal Interaction and their Base Performance.

Total Rating = Internal Interaction + Base Performance.

$$x = Ax + d \quad \text{(In matrix)}$$

by rearranging,

$$x = (I - A)^{-1} d$$

where,

- I is $n \times n$ Identity Matrix
- Substracting Ax from both sides,
  $(I - A)x = d$.
- $(I - A)$ is non-singular

This calculates the Total Rating required for each player to meet their bar performance while also accounting strength of other players in the system.

Code -

```
clc;
clear;


% 1.In this case study, we treat players as interdependent entities[cite: 3].
players = {'Alice', 'Bob', 'Charlie'};

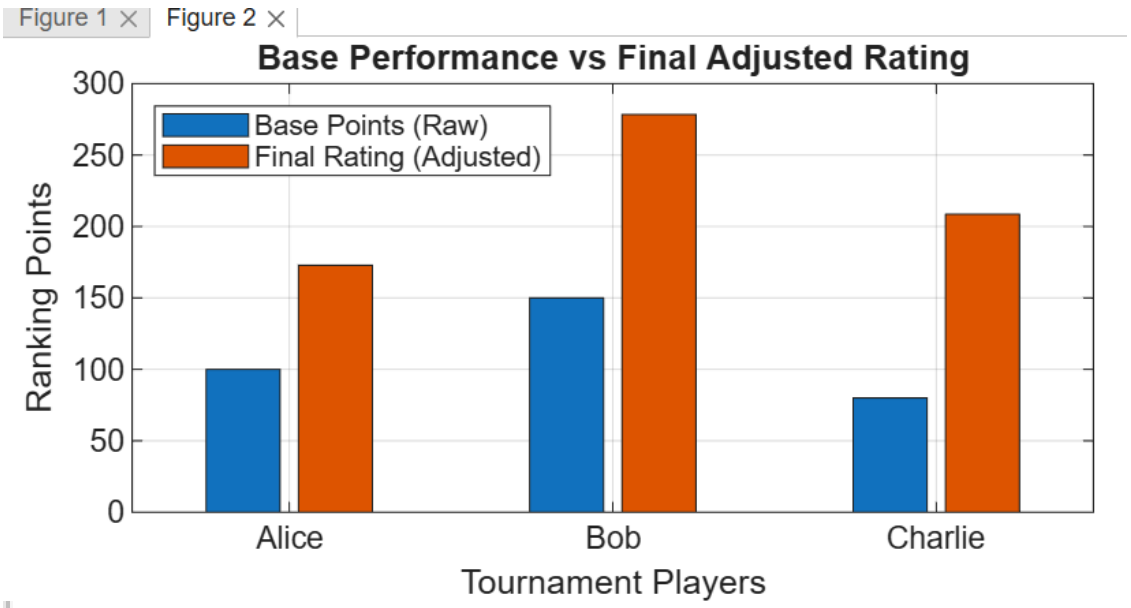
% 2. Interaction (Technology) Matrix A
% a_ij represents the "strength" player i derives from player j.
% If Alice (1) beats Bob (2) frequently, A(1,2) is higher.
A = [0.20, 0.10, 0.05;  % Alice's dependency on others
     0.30, 0.20, 0.10;  % Bob's dependency on others
     0.10, 0.25, 0.20]; % Charlie's dependency on others [cite: 38, 51-53]

% 3. Base Performance Vector d (Replaces Final Demand)
% The raw wins or points scored by each player [cite: 39, 57-59].
d = [100; % Alice's base points
     150; % Bob's base points
     80]; % Charlie's base points

% 4. Ranking Calculation (Leontief Equation)
% Using the balance equation: (I - A)x = d [cite: 31, 64]
I = eye(size(A));      % Identity matrix [cite: 32, 63]
x = (I - A) \ d;       % Final Ratings (Total Output) [cite: 34, 64]

% 5. Display Numerical Results
fprintf('Final Adjusted Rankings:\n');
for i = 1:length(players)
    fprintf('%s: %.2f Rating Points\n', players{i}, x(i));
end

% 6. Visualization
figure('Color', 'w');
hBar = bar([d, x]);
set(gca, 'XTickLabel', players, 'FontSize', 11);
ylabel('Ranking Points');
xlabel('Tournament Players');
title('Base Performance vs Final Adjusted Rating');
legend('Base Points (Raw)', 'Final Rating (Adjusted)', 'Location', 'northwest');
grid on; [cite: 68-74]
```

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Final Adjusted Rankings:
Alice: 172.84 Rating Points
Bob: 278.39 Rating Points
Charlie: 208.60 Rating Points
```

Conclusion

The application of the Leontief Input-Output model to tournament rankings demonstrates that player performance is not an isolated metric but a result of a complex, interdependent system. By treating match results as "intermediate inputs" and raw wins as "final demand," the matrix model successfully adjusts a player’s score based on the quality of their competition.

The resulting data confirms that a player's Total Rating (x) must exceed their Base Performance (d) to account for the mathematical weight of their schedule. This approach ensures a fair, objective leaderboard where "backbone" players—those who consistently face and beat high-tier opponents—are correctly identified as the most central figures in the tournament's ecosystem.